

MICHAEL FANTINATO

MSc Computer Science Student | Full Stack Developer & Automation Engineer

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EDUCATION

•IT University of Copenhagen

MSc in Computer Science

Aug 2025 – Present
Copenhagen, Denmark

•University of Padua

BSc in Computer Science

Sep 2021 – Jul 2025
Padua, Italy

WORK EXPERIENCE

•DEVESS S.R.L.

Full Stack Developer

Apr 2024 – Jul 2025
Piombino Dese, Italy

- Designed and implemented a Flutter application for a startup, creating a "flow regulator" system for manufacturing companies to manage and optimize production efficiency.
- Developed features enabling workers and managers to identify production bottlenecks and propose targeted improvements to the assembly line.
- Assumed managerial responsibilities including cost estimation, task breakdown/assignment, and the application of Agile methodologies to guide the development process.

•F.C. Servizi

Full Stack Developer

Oct 2021 – Apr 2024
Piombino Dese, Italy

- Addressed operational challenges by developing a custom Flutter application to track incoming components for industrial machinery.
- Engineered a system providing real-time details on material location, movement history, and quantities, drastically reducing manual search times.
- Improved physical logistics by redesigning warehouse layouts and optimizing loading/unloading processes, resulting in higher overall efficiency.

PROJECTS

•AI-Powered Production Optimization (Bachelor's Thesis)

Microservices Architecture & LLM Integration

May 2025 – Jul 2025

- Designed a software platform to optimize industrial production lines, featuring a microservices architecture with a Python/FastAPI backend and React/TypeScript frontend.
- Integrated an Anthropic Claude LLM using LangChain to analyze production data and generate actionable recommendations.
- Engineered a custom 'Model Context Protocol' (MCP) server as a secure API gateway, allowing the LLM to safely query and execute operations on a MongoDB database.
- Built a data analytics service with Pandas to process raw production data, calculate KPIs, and detect anomalies.

•RaceUp - Formula SAE Team (University of Padua)

Control System Manager (Driverless Division)

Oct 2023 – Aug 2025

- Collaborated within a high-performance multidisciplinary team of **120+ students**, coordinating specifically with perception and vehicle dynamics departments to integrate the control stack.
- Led the implementation of high-level (Pure Pursuit, Stanley) and low-level (PID) controllers for an autonomous racing vehicle, working closely with researchers to optimize lap times.
- Laid the foundation for the autonomous driving system by configuring camera hardware and developing the initial path-planning nodes.
- Contributed to the mechanical and electrical construction of a dedicated test vehicle, enabling data collection and validation of algorithms, sensors, and ECUs under track conditions.

•Corporate Chatbot - Software Engineering Project

RAG System for Internal Documentation

Oct 2024 – May 2025

- Developed a corporate chatbot in collaboration with Azzurro Digitale inside a team of 7 people, designed to answer company-specific queries by retrieving data from internal documents.
- Engineered integrations with external applications (Confluence, Jira, GitHub) to query proprietary data.
- Built the full-stack web application using a Python backend and an Angular frontend.

•Industrial Automation Machine

Falegnameria Tommasello | Automation Design

Apr 2024 – Sep 2024

- Designed the complete electrical automation system for a custom industrial parts processing machine (moving, cutting, drilling).
- Selected and integrated sensors/actuators and developed final control software on PLCs to manage the machine's full operational cycle.

TECHNICAL SKILLS

Languages: Python, Dart, JavaScript, TypeScript, C++, Java, SQL, HTML/CSS, C, Scala

Frameworks & Libraries: Flutter, FastAPI, Spring Boot, React, Angular, LangChain, Pandas, NumPy

Infrastructure & Tools: Docker, Git, Fusion 360, 3D Experience, MongoDB, PostgreSQL, PLCs, Agile Methodologies, Arduino

Concepts: Microservices, LLM Integration (RAG), Industrial Automation, Autonomous Control Systems (PID, Stanley)

Spoken Languages: English (Advanced - TOEFL: 96/120), Italian (Native)